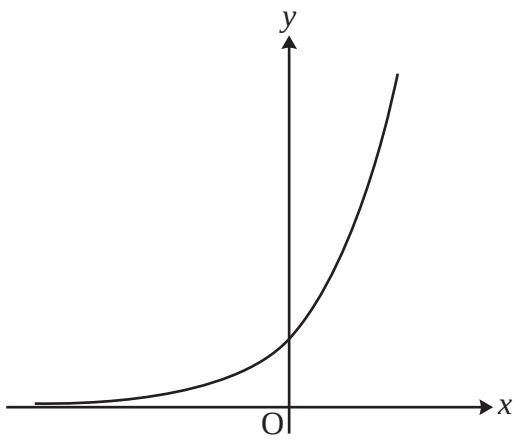
**5** **(a)** The diagram shows the curve $y = e^x$.




On the axes in the Printed Answer Booklet, sketch graphs of

**(i)** $\frac{dy}{dx}$ against $x$, **[1]**

**(ii)** $\frac{dy}{dx}$ against $y$. **[2]**

**(b)** Wolves were introduced to Yellowstone National Park in 1995.

The population of wolves, $y$, is modelled by the equation

$y = Ae^{kt}$,

where $A$ and $k$ are constants and $t$ is the number of years after 1995.

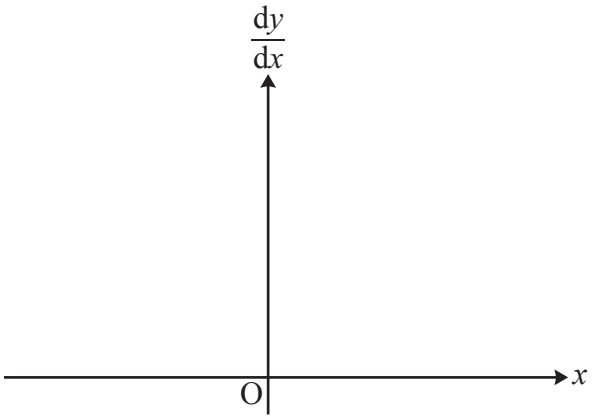
**(i)** Give a reason why this model might be suitable for the population of wolves. **[1]**

**(ii)** When $t = 0$, $y = 21$ and when $t = 1$, $y = 51$.

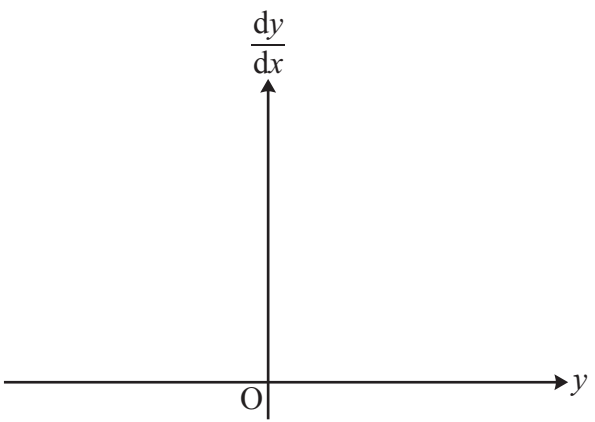
Find values of $A$ and $k$ consistent with the data. **[3]**

**(iii)** Give a reason why the model will not be a good predictor of wolf populations many years after 1995. **[1]**

**5(a)(i)**




**5(a)(ii)**



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|---|---|
| **5(b)(i)** | |
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| **5(b)(ii)** | |
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| | |
| | $A =$ |
| | $k =$ |
| **5(b)(iii)** | |
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